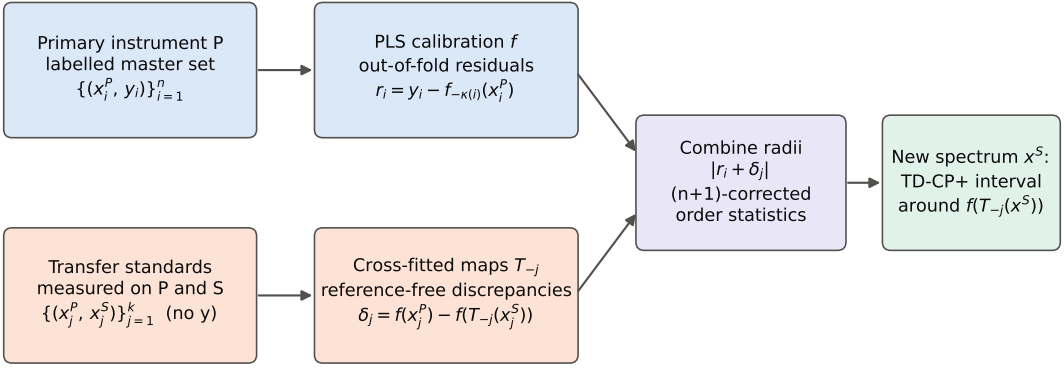$$y - f(T(x^S)) = [y - f(x^P)] + [f(x^P) - f(T(x^S))] = r + \delta$$